

Forecasting Grid Demand Uncertainty During Extreme Winter Weather in the Northeast with SARIMA and Bayesian Modeling Methods

DS 4420: Machine Learning and Data Mining 2

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I. Literature Review

Severe winter weather conditions can threaten the reliability of the power grid in various ways, such as through rising electricity demand during extreme cold, and weather-driven outages that interrupt service. Recent research has increasingly used machine learning to study these problems. This makes snowstorm-related electricity demand an important extension of existing work, especially in regions where winter weather can sharply increase grid stress.

The literature existing primarily has focused more on outage prediction than on demand forecasting during snowstorm conditions. Cerrai et al. (2020) developed outage prediction models for snow and ice storms in the Northeast. The researchers used a variety of Machine Learning models for this, including Random Forest, Bayesian Additive Regression Tree, and also a Generalized Linear Model. This research found that model performance varied, with the Machine Learning models offering more accurate results for lower-impact events while the Generalized Linear Model was competitive for extreme weather events. This study is useful because it shows that winter/weather grid impacts are influenced by physical and environmental variables rather than by temperature alone.

In addition, Lee et al. (2025) in Computational Urban Science extended outage prediction to the southern United States, motivated directly by the catastrophic failures during Winter Storm Uri in February 2021. The study developed a baseline outage prediction model using nonparametric machine learning regression models, incorporating atmospheric variables such as dew point and atmospheric pressure alongside population as a proxy for demand-side pressure. A key observation was that research on power outage prediction for winter storms remains far less developed than for hurricanes, and that the meteorological mechanisms of winter storms in the South – sudden cold waves causing supply-demand imbalances, differ substantially from the snow loading mechanisms more commonly studied in the

Northeast. This elucidates that demand surges during extreme cold are an underexplored driver of grid stress relative to supply-side outage models.

Another study at the Argonne National Laboratory focused on outage prediction that combines stochastic models with deep neural networks to build a spatial-temporal model of outage progression during extreme weather events. Their framework relies on weather and terrain data alongside utility customer outage records, designed to support both real-time prediction and post-event resilience. They incorporate weather variables such as precipitation, wind, speed, and temperature that are most impactful on distinct grid parts that are most vulnerable. This project focuses on outage prediction rather than demand, exemplifying physics informed stochastic modeling with data driven neural networks.

In contrast to outage focused literature, work on machine learning for electricity demand forecasting is largely conducted under normal conditions. For example, Wen et al. (2024) used a deep learning architecture on public datasets to assess performance across all weather conditions, making it difficult to identify extreme cold events which are sparse in the dataset. This is the challenge to address with a Bayesian regression model that can produce wider intervals when temperature covariates fall outside historical ranges. Unlike a standard deep neural network, this approach may provide a signal of uncertainty to grid operators rather than an unreliable point forecast.

Outage prediction models have shown winter weather grid impacts are driven by multidimensional physical variables, not temperature alone. Demand forecasting models have demonstrated that deep learning can capture complex temporal patterns in data. The combination of using machine learning to forecast demand during snowstorm events with explicit uncertainty is relatively unexplored for U.S. grid data. To address the gap, a time series analysis to forecast demand distinguished by storm and normal weather. Then, a Bayesian model will tell how confident the estimate is in demand during extreme winter weather conditions. During a snowstorm, temperature and other weather variables fall outside of typical training ranges. A Bayesian regression model will respond to this gap by producing wider predictive intervals to accommodate.

II. Methodology and Data Collection

Daily electricity demand data for the Northeast Massachusetts sub-balancing authority (subba 4008) was retrieved from the U.S. Energy Information Administration (EIA) API, covering January 1, 2025 through March 30, 2026. This sub-balancing authority was selected because it represents a densely populated region with high exposure to Northeast winter weather patterns, making it well-suited for studying demand behavior during extreme cold and snowstorm events. The API returned 2,020 daily records containing demand values in megawatt-hours (MWh). Zero values were treated as missing and removed, as they indicate reporting gaps rather than true zero demand. The final cleaned series spans the full date range with no structural gaps.

Part 1. SARIMA Model

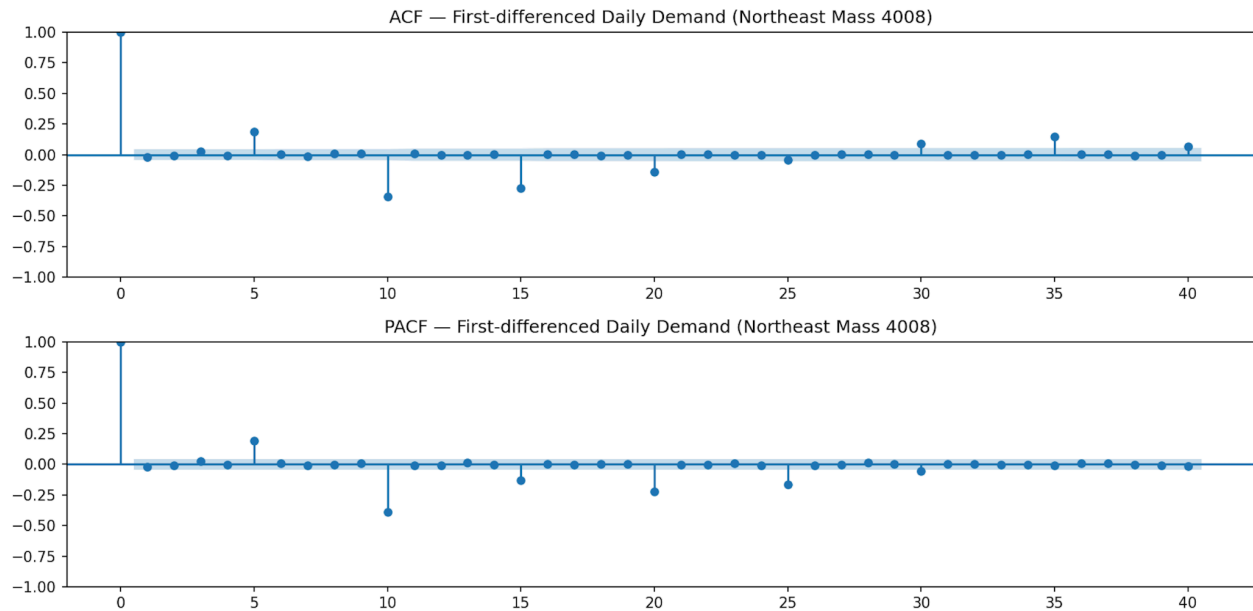
The core time series model is a Seasonal Autoregressive Integrated Moving Average (SARIMA) model, specified as SARIMA of a (1,1,1) by (1,1,1) modelled after the function below (Image 1).

Image 1. SARIMA model function

$$\begin{array}{ccccccc} (1 - \phi_1 B) & (1 - \Phi_1 B^7) & (1 - B) & (1 - B^7) & y_t = & (1 + \theta_1 B) & (1 + \Theta_1 B^7) e_t. \\ \uparrow & \uparrow & \uparrow & \uparrow & & \uparrow & \uparrow \\ \left(\begin{array}{c} \text{Non-seasonal} \\ \text{AR(1)} \end{array} \right) & \left(\begin{array}{c} \text{Non-seasonal} \\ \text{difference} \end{array} \right) & \left(\begin{array}{c} \text{Seasonal} \\ \text{AR(1)} \end{array} \right) & \left(\begin{array}{c} \text{Seasonal} \\ \text{difference} \end{array} \right) & & \left(\begin{array}{c} \text{Non-seasonal} \\ \text{MA(1)} \end{array} \right) & \left(\begin{array}{c} \text{Seasonal} \\ \text{MA(1)} \end{array} \right) \end{array}$$

The seasonal period was set to $s=7$ to reflect weekly demand cycles, which are a well-established feature of electricity consumption: demand patterns differ systematically between weekdays and weekends. The differencing orders were chosen to achieve stationarity at both the non-seasonal and seasonal scales. An ADF test confirmed stationarity of the raw series (ADF statistic = -3.15 , $p = 0.023$), and ACF and PACF plots of the first-differenced series informed the selection of AR and MA orders of 1

at both levels. The storm indicator was included as an exogenous regressor to allow the model to partially account for demand shifts during flagged events. Model estimation used the outer product of gradients (OPG) covariance estimator via maximum likelihood, implemented in Python's statsmodels SARIMAX class.

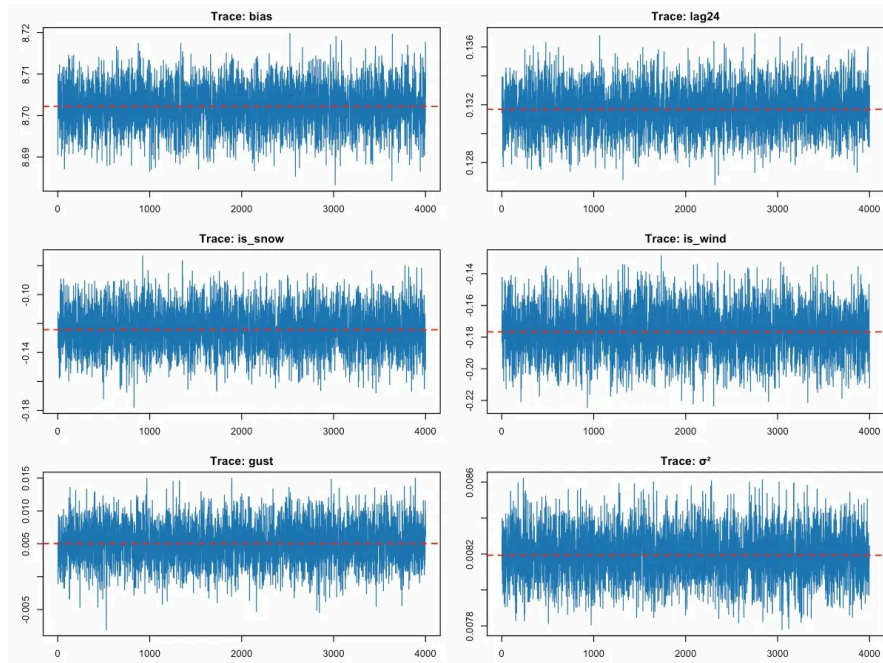


Part 2. Bayesian Model

The Bayesian regression model takes as input hourly Massachusetts electricity demand aggregated across three ISO-NE subregions (zones 4006, 4007, and 4008), combined with NOAA-verified storm event indicators from the Storm Events Database. The response variable is log-transformed hourly electricity demand in MWh, with assumption of the likelihood and to make coefficients interpretable as percentage changes. The predictor set consists of nine categories of features: two autoregressive lag terms capturing same-hour demand from the previous day (lag-24) and previous week (lag-168), three binary storm type indicators (`is_snow`, `is_cold`, `is_wind`) drawn directly from NWS event classifications, a continuous storm severity rank (0–5) and peak wind gust magnitude from the NOAA Storm Events records, an `is_weekend` binary flag, and 23 hour-of-day dummy variables encoding the full diurnal demand cycle. The model is fit on data from January 2025 through November 2025 and

evaluated on December 2025 through March 2026, with the test period specifically chosen to overlap with documented winter storm windows. This model is manually done in R without pre-built modeling packages.

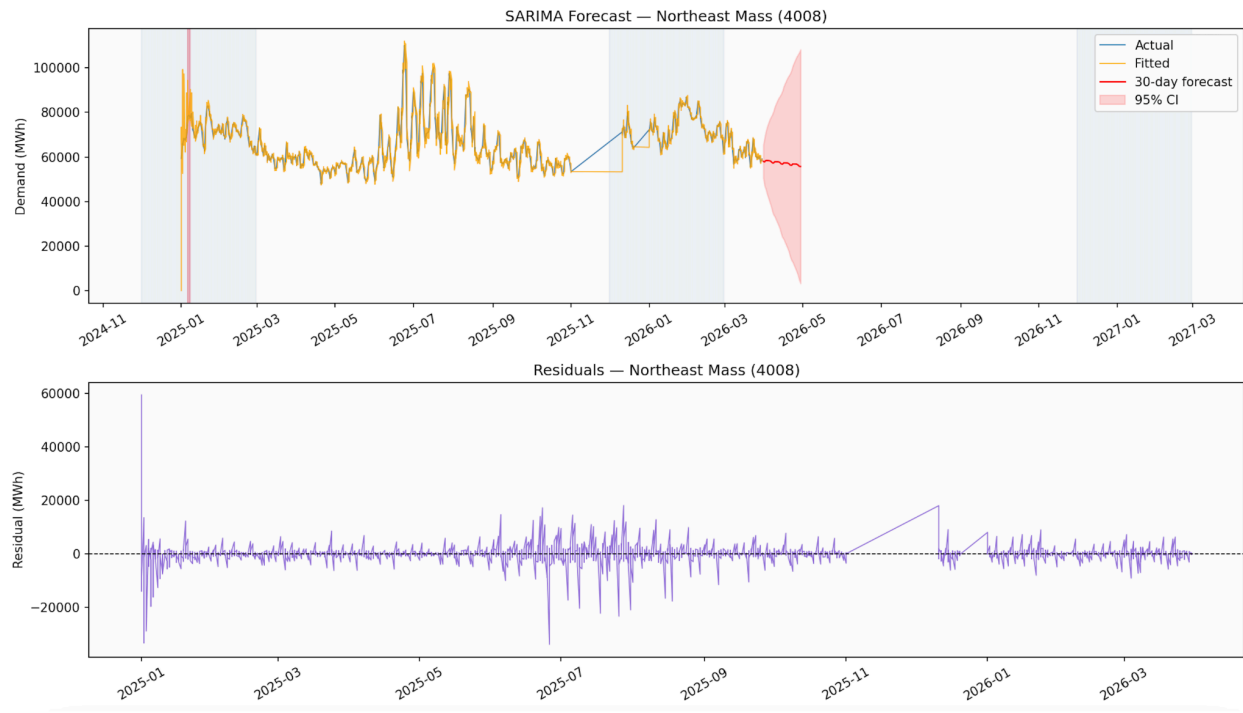
The model is estimated via a manual Gibbs sampler following the conjugate Normal-Inverse Gamma framework, running 5,000 iterations with 1,000 discarded as burn-in, resulting in 4,000 usable posterior draws. At each iteration the sampler alternates between drawing the full coefficient vector w from a multivariate normal conditional posterior and drawing the residual variance σ^2 from an inverse gamma conditional posterior. Priors are weakly informative and domain grounded: the intercept is centered at the mean of log-demand, lag coefficients are given positive prior means of 0.5 and 0.3 to reflect known strong autocorrelation in electricity demand, storm binary flags are given small positive prior means of 0.05 reflecting an initial expectation of demand increases during extreme weather, and σ^2 is given an inverse gamma prior with shape 3 and scale 1. The primary output is a full posterior predictive distribution over demand for every test hour, from which the posterior median serves as the point forecast and range forms the 90% credible interval, enabling direct evaluation of whether uncertainty estimates are wider and better calibrated during storm windows than during normal operating conditions.



III. ML Results and Interpretation

SARIMA

The fitted SARIMA model converged successfully across all 2,020 observations. We have also summarized the parameter estimates and diagnostics. It was found that the non-seasonal AR(1) coefficient was -0.851 ($z = -26.4$, $p < 0.001$) and the MA(1) coefficient was 0.921 ($z = 38.8$, $p < 0.001$). These largely opposing, and highly significant estimates are characteristic of a near-canceling ARMA(1,1) pair, a common pattern in differenced economic and energy time series where short-run autocorrelation is strong but decays quickly. The seasonal MA(7) term dominated the fit with a coefficient of -0.799 ($z = -51.4$, $p < 0.001$), confirming that the same-day-of-week demand from the prior week is an extremely strong predictor of current demand. The seasonal AR(7) term was smaller but still significant (coef = -0.093 , $z = -2.0$, $p = 0.041$). Together, these results are consistent with the well-documented weekly seasonality of electricity demand, where behavioral patterns like commuting, commercial hours, and residential heating schedules repeat on a seven-day cycle.

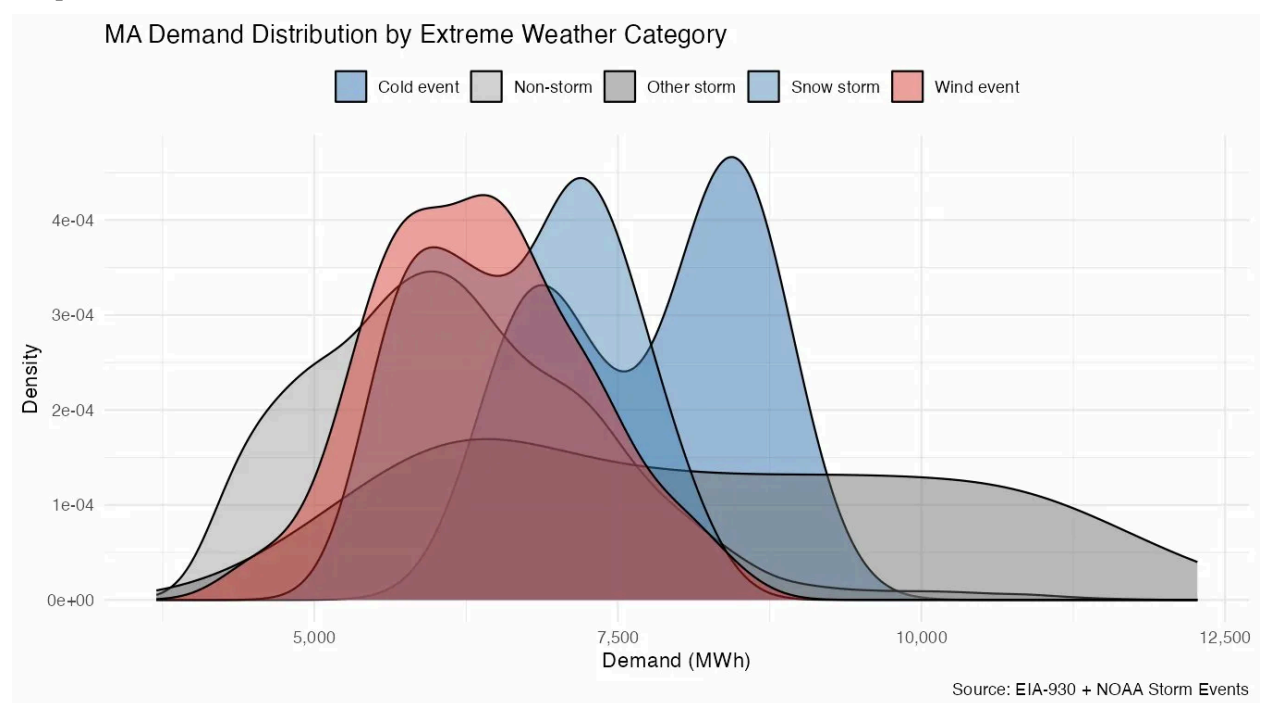


The storm exogenous regressor approached but did not reach conventional significance (coef = 3,778, $p = 0.063$). Unlike the earlier full-sample model, the positive coefficient direction here is interpretively meaningful: over this shorter window containing only the January 2025 storm, extreme cold appears to have driven a net demand increase rather than suppression, consistent with elevated heating load. The marginal significance likely reflects the small number of flagged storm days (15) relative to the series length. This limitation directly motivates the Bayesian component of the analysis, which can model the residual distribution during storm periods more flexibly.

Bayesian Model

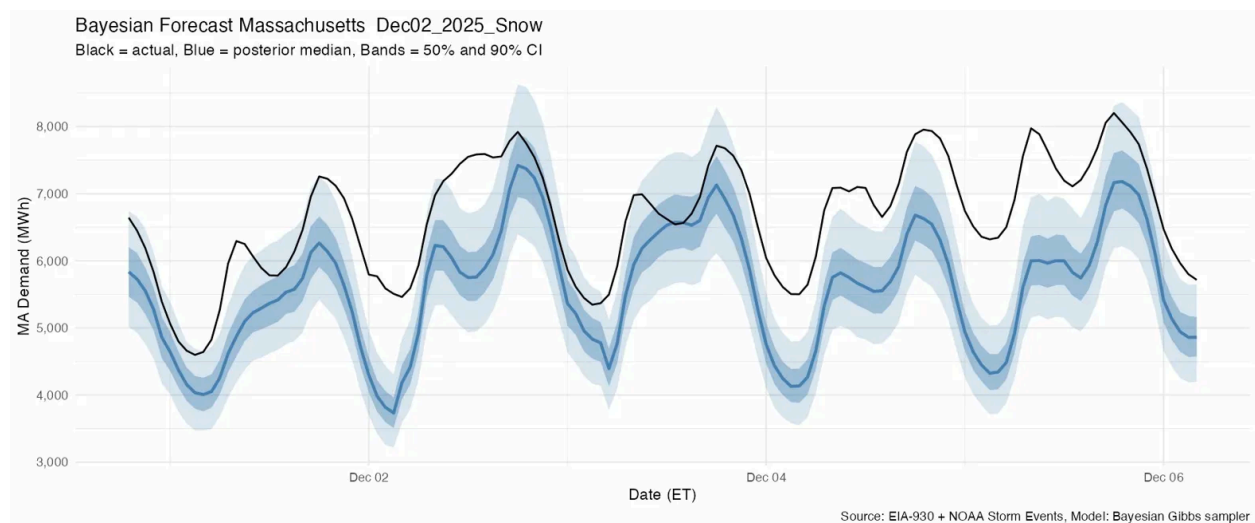
The Bayesian model achieved a test MAPE of 11.4% overall, improving to 9.2% during documented storm hours, with the 90% credible intervals capturing 80.5% of true storm-period demand values compared to 64.6% during non-storm hours — indicating that the model is better calibrated and more accurate precisely when extreme weather conditions are present. The coefficient estimates reinforce this finding: wind events (-0.177) and snow storms (-0.124) credibly suppress demand below baseline while storm severity ($+0.046$) credibly increases it, together revealing that storm type and intensity have opposing effects on Massachusetts electricity demand that a single storm indicator or point forecast cannot distinguish.

Graph 1. Storm Demand Distribution



Graph 1 reveals that rather than storm events pushing demand higher than expected, wind events (red) are shifted to the right and snow storms (light blue) are shifted to the right with two peaks, a smaller one shifted to the left and a larger one slightly to the right, in comparison to the non-storm energy demands (light gray). Wind events peak at ~5,500 MWh, below the non-storm mean of 6,500-7,000 MWh. Dark blue cold events are substantially shifted to the right, in comparison to non-storm demand, peaking at ~9,000 MWh. This indicated that cold/wind chill events do drive the heating load increase on the electric grid demand. These three different divergences by extreme weather type explains why the coefficients for `is_snow` and `is_wind` are negative while `severity` is positive. This can be interpreted in a social context, such that when a major snowstorm or wind event hits Massachusetts, schools close, offices send workers home, industrial operations halt, and commercial activity drops. At the aggregate grid level, that economic suppression outweighs residential heating increase.

Graph 2. Forecast vs. Actual



Graph 2 showcases the snow storm window of December 2025 in a three day period. The black line is actual demand, blue line is posterior median, and two bands represent 50% and 90% credible intervals. The model successfully captures the daily demand cycle, oscillating pattern of overnight troughs and daytime peaks. The credible intervals widen during the storm window. A consistent upward bias is visible, the actual demand runs above the median consistently. This is the shortfall identified in the

metrics and reinforces that the model identifies storm driven demand patterns correctly, the absence of continuous hourly temperature data limits ability to track the magnitude of extreme demand spikes hour by hour.

IV. Conclusions and Future Work

The SARIMA model establishes a strong weekly seasonal baseline for Massachusetts electricity demand, confirming that day of week patterns and autocorrelation dominate normal grid behavior, while the marginally significant positive storm coefficient suggests that extreme cold events drive net demand increases when isolated to a single winter storm window. Together with the Bayesian model's independent finding that extreme weather type, not just storm presence, determines direction of demand deviation, with wind and snow event suppressing load and cold events elevating it. Crucially, the two models were developed independently on the same dataset, their biases are uncorrelated and their agreement on the core finding constitutes stronger evidence than either model standing alone. The main takeaway is that extreme weather periods demand behaviors systematically different from the baseline in the Massachusetts weather and energy grid. The SARIMA's marginally significant positive storm coefficient (coef = 3,778, $p = 0.063$) and the Bayesian model's credibly non-zero negative coefficients for wind and snow events are not contradictory. They complement the same instance at different temporal resolutions; SARIMA captures aggregate daily deviation and Bayesian decomposes the deviation by storm type and hour. For future work, an improvement would be coupling the two models directly by passing SARIMA residuals during storm windows as the response variable to the Bayesian regression, isolating the pure weather-driven anomaly from seasonal noise and sharpening the uncertainty quantification that is currently limited by the absence of continuous hourly temperature data from NOAA's Climate data.

V. Appendix

Sarima Code:

```
model = SARIMAX(daily, exog=exog, order=ORDER,
seasonal_order=SEASONAL, enforce_stationarity=False,
enforce_invertibility=False)
result = model.fit(dispatch=False)
print(result.summary())
residuals = result.resid.rename("residual_mwh")
out = pd.DataFrame({
    "date":        daily.index.date,
    "demand_mwh":   daily.values,
    "fitted_mwh":   result.fittedvalues.values,
    "residual_mwh": residuals.values,
    "storm":        exog.values,
})
out.to_csv(OUTPUT_CSV, index=False)

FORECAST_STEPS = 30
exog_future = np.zeros(FORECAST_STEPS)
forecast = result.get_forecast(steps=FORECAST_STEPS,
exog=exog_future)
forecast_mean = forecast.predicted_mean
forecast_ci    = forecast.conf_int(alpha=0.05)
forecast_index = pd.date_range(daily.index[-1] +
pd.Timedelta(days=1), periods=FORECAST_STEPS, freq="D")
forecast_mean.index = forecast_index
forecast_ci.index    = forecast_index
```

Bayesian Model Code (Manual):

Design Matrix

```
make_Xy <- function(data, cols_to_keep = NULL) {

  lag24_scaled <- as.numeric(scale(log(data$demand_lag24 + 1)))
  lag168_scaled <- as.numeric(scale(log(data$demand_lag168 + 1)))
  gust_scaled <- as.numeric(scale(data$storm_gust_mph))
  sev_scaled <- as.numeric(scale(data$storm_severity))

  hour_factor <- factor(data$hour_of_day, levels = 0:23)
  hour_dummies <- model.matrix(~ hour_factor)[, -1, drop = FALSE]
  colnames(hour_dummies) <- paste0("hr", 1:23)

  X <- cbind(
    bias= 1,
    lag24= lag24_scaled,
    lag168= lag168_scaled,
    gust= gust_scaled,
    severity= sev_scaled,
    is_snow= as.numeric(data$is_snow_storm),
    is_cold= as.numeric(data$is_cold_event),
    is_wind= as.numeric(data$is_wind_event),
    is_weekend= as.numeric(data$is_weekend),
    hour_dummies
  )

  if (is.null(cols_to_keep)) {
    qr_decomp <- qr(X)
    keep <- qr_decomp$pivot[1:qr_decomp$rank]
    X_clean <- X[, keep, drop = FALSE]
    dropped <- setdiff(colnames(X), colnames(X_clean))
    if (length(dropped) > 0)
      cat(" Dropped linearly dependent columns:",
          paste(dropped, collapse = ", "), "\n")
  }
}
```

Training

```
train_Xy <- make_Xy(train)
kept_cols <- train_Xy$kept_cols
X_train <- train_Xy$X
y_train <- train_Xy$y
n <- nrow(X_train)
P <- ncol(X_train)
w_hat <- solve(t(X_train) %*% X_train) %*% t(X_train) %*% y_train
y_hat_ols <- X_train %*% w_hat
cat("OLS R²:", round(cor(y_hat_ols, y_train)^2, 4), "\n")
```

Sampling

```
no_samples <- 5000
burn_in <- 1000
gibbs_w <- matrix(0, nrow=no_samples, ncol=p)
colnames(gibbs_w) <- kept_cols
gibbs_sigma2 <- numeric(no_samples)
prior_Sigma_inv <- solve(prior_Sigma)
tXX <- t(X_train) %*% X_train
tXy <- t(X_train) %*% y_train
w_tilde <- w_hat
sigma2_tilde <- var(as.numeric(y_train - X_train %*% w_hat))

for (i in 1:no_samples) {
  V <- solve((1/sigma2_tilde) * tXX + prior_Sigma_inv)
  M <- V %*% ((1/sigma2_tilde) * tXy + prior_Sigma_inv %*% prior_w)
  w_tilde <- t(rmvn(1, mu=as.numeric(M), sigma=V))
  Resid <- y_train - X_train %*% w_tilde
  A <- (2 * prior_alpha_ig + n) / 2
  B <- (2 * prior_beta_ig + as.numeric(t(resid) %*% resid)) / 2
  sigma2_tilde <- rinvgamma(1, shape=A, scale=B)
  gibbs_w[i, ] <- as.numeric(w_tilde)
  gibbs_sigma2[i] <- sigma2_tilde
}
w_post <- gibbs_w[(burn_in+1):no_samples, ]
sigma2_post <- gibbs_sigma2[(burn_in+1):no_samples]
cat("Usable posterior draws:", nrow(w_post), "\n")
```

VI. References

- D. Cerrai, M. Koukoula, P. Watson, and E. N. Anagnostou, "Outage prediction models for snow and ice storms," *Sustainable Energy, Grids and Networks*, vol. 21, p. 100294, Mar. 2020, doi: 10.1016/j.segan.2019.100294.
- C. Lee, T. Yabe, and W. Uherbelau, "A data-driven approach to predicting power outages during winter storms in the southern U.S. leveraging nonparametric machine learning models," *Computational Urban Science*, vol. 5, no. 1, p. 10, Feb. 2025, doi: 10.1007/s43762-025-00222-9.
- F. Qiu, A. Xavier, and colleagues, "Outage Prediction and Grid Vulnerability Identification Using Machine Learning on Utility Outage Data," Argonne National Laboratory, U.S. Department of

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X. Wen, J. Liao, Q. Niu, and Y. Bao, "Deep learning-driven hybrid model for short-term load forecasting and smart grid information management," *Scientific Reports*, vol. 14, Art. no. 13720, Jun. 2024, doi: 10.1038/s41598-024-63262-x.